Template Document

Academic Support Agent

February 19, 2025

Dear Application Committee,

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In econometrics and genomics, the establishment of statistical inferences is often distorted by confounding variables that cause pseudo-correlation and lead to distortion of causality. The doubly debiased lasso method, on the other hand, aims to efficiently estimate implicitly confounding variable models in the high-dimensional case by simultaneously removing the bias caused by confounding variables and regularization, resulting in unbiased estimates.

Partially linear model is a kind of semiparametric model, including both linear regression and nonlinear functions in itself. Using a partially linear model as a framework allows for the convenient introduction of both nonlinear trends and confounding variables, which is consistent with the problem of large-scale nonparametric or semiparametric estimation under confounding in realistic data analysis.

Sincerely,